Course Code : ECT 453-3 **KQLR/RS – 24 / 1680**

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

MACHINE LEARNING

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory and carry marks as indicated.
(2) Assume suitable data wherever necessary.
(3) Illustrate your answer with neat sketches.
(4) Use of Non – programmable calculator is permitted.

1. (a) Feature x is normally distributed, with mean = 3 and std. dev. = 2. Find $P(-3 < x < -2)$. 3(CO2)

(b) If $p(x) = 2x/9$ for $0 < x < 3$, and $p(x) = 0$ otherwise, determine the value of $P(0 < x < 1)$. 4(CO1,2)

(c) A classifier has a 30 percent error rate. What is the probability that exactly three errors will be made in classifying 10 samples ? 3(CO1,2)

2. (a) Write a Python script to read a csv file named "Nagpur.csv" and convert the read data into data frame. Print the different statistical parameters using one statement and correlation matrix of the loaded data set using heatmap. 4(CO3,4)

(b) Write a Python script to read a csv file named "stock_price.csv" and convert the read data into data frame. Look for categorical features and encode them using one hot encoding. Visualize features using pairplot. Apply Principal Component Analysis to cover a variance of 95%. 6(CO3,4)

3. (a) Determine the optimal least risk decision boundaries for a feature x is normally distributed for class A with mean = 0 and Std. Dev. = 1 and normally distributed for class B with mean = 1 and Std. Dev. = 2. If the loss for misclassifying a sample that is really from class A is INR 10000, the loss for class B is INR 50, and $P(A) = P(B)/100$. 6(CO2,4)

(b) Write a Python script to predict the next day rainfall in Nagpur using "rainfall_Nag.csv". Apply any two techniques with cross validation and print the error metrics. 4(CO3,4)

4. (a) Apply K-NN technique for k = 3 using city block distance to classify a sample with a feature vector (1, 1) if samples from class A are at (3, 0), (4, 1) and (3, 2) and samples from class B are at (1, –1) and (1, –1·5). 5(CO2,4)

(b) Write a Python script to split the data present in X (features) and Y (targets) into 80% train and 20% test set. Use Support Vector Machine without regularization and with L1 and L2 regularization. Include the statement to plot the confusion matrix and print the Precision and Recall values. 5(CO3,4)

5. (a) Perform a K-means clustering. Set k = 2 and use the first two samples in the list as the initial samples.

	1	2	3	4	5	6	7
x	0	0·5	0	2	2·5	6	7
y	0	0	2	2	8	3	3

Illustrate the distance matrices and the dendrogram. 7(CO2,4)

(b) Explain Back-propagation algorithm with an example. 3(CO2)

6. (a) A Drone company owner has come to you for automating the crowd management at a huge social gathering using Machine learning. As a machine learning system designer, explain the different components required for the system with block diagram. Which features and how many will you choose ? What are your challenges and how will you overcome them ? 7(CO5)

(b) Consider a scenario that Machine Learning is being implemented in Indian cities for video surveillance for detecting antisocial activities and prevention of crime. Decide and justify that whether it is better for the society or breach of privacy. 3(CO1,2)